

Secure Two-Factor Authentication for IoT Device

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Abstract—The development of IoT has penetrated various sectors. The development of IoT devices continues to increase and is predicted to reach 75 billion by 2025. However, the development of IoT devices is not followed by security developments. Therefore, IoT devices can become gateways for cyber attacks, including brute force and sniffing attacks. Authentication mechanisms can be used to ward off attacks. However, the implementation of authentication mechanisms on IoT devices is challenging. IoT devices are dominated by constraint devices that have limited computing. Thus, conventional authentication mechanisms are not suitable for use. Two-factor authentication using RFID and fingerprint can be a solution in providing an authentication mechanism. Previous studies have proposed a two-factor authentication mechanism using RFID and fingerprint. However, previous research did not pay attention to message exchange security issues and did not provide mutual authentication. This research proposes a secure mutual authentication protocol using two-factor RFID and fingerprint using MQTT protocol. Two processes support the authentication process: the registration process and authentication. The proposed protocol is tested based on biometric security by measuring the false acceptance rate (FAR) and false rejection rate (FRR) on the fingerprint, measuring brute force attacks, and measuring sniffing attacks. The test results obtained the most optimal FAR and FRR at the 80% threshold. Then the equal error rate (ERR) on FAR and FRR is around 59.5%. Then, testing brute force and sniffing attacks found that the proposed protocol is resistant to both attacks.

Index Terms—IoT device, Mutual authentication, Two-factor authentication, Constraint device, MQTT

I. INTRODUCTION

The development of internet-based intelligent devices, or the internet of things (IoT), is multiplying. The development of IoT has penetrated various important sectors in life, such as health, transportation, education, and food supply [1]–[6]. IoT has the concept of a collection of many objects or devices that can communicate with each other by using the internet. IoT devices will reach 75 billion devices by 2025 [1], [3]. Of course, this is a perfect opportunity to develop IoT devices.

The IoT has four layers: physical, network, server, and application layers [4]. The physical device provides and executes data from and to the server through the network.

The physical layer is where the physical devices, such as sensors and actuators, are located in the environment. These devices generally consist of a small controller such as a micro-controller. The use of micro-controllers has an excess of low power usage and dimension [3], [7]. These make the devices can mobilize quickly. The physical device needs a communication protocol to communicate with the server. The lightweight communication protocol, such as message queuing telemetry transport (MQTT), is undoubtedly a mainstay [2], [4], [5]. The MQTT computation cost is light. However, the lightweight-oriented communication protocol focused on only building fast and low computation costs. The security issues have been concerning for the IoT devices that use lightweight-oriented communication protocol.

The authentication issue is one of the vulnerabilities caused by the lightweight-oriented communication protocol [8]–[17]. Besides, the authentication mechanism is crucial for every computer network system. Authentication allows the legitimate user to use the services provided by the server. Thanks to the authentication mechanism, the server can choose the legitimate data source for further process. Therefore, the data received by the server are the trusted data. However, implementing an authentication mechanism has been challenging for IoT physical devices [18]–[22]. The challenge happens due to the condition and the capability of the IoT physical devices [3], [7], [22]–[26]. Implementation of conventional authentication mechanisms is not suitable for use on constraint devices.

The authentication mechanism required the property to evaluate. The property can be from three sources: text-based, device-based, and biometric-based. The proper choice of property source can improve security. The use of text-based sources is weak [27]. The text-based sources are vulnerable to brute force attacks, such as dictionary attacks [27]. The uses of either device-based and biometric-based sources are better than text-based [27]–[30]. Radio-frequency identifier (RFID) is one of the device-based sources. The RFID is a typical device to provide authentication properties. The fingerprint is one of the biometric-based sources. Recently, fingerprint is commonly used to provide authentication [28], [29]. Combining those

two sources can increase security [27]–[30]. The research conducted to implement those combinations was done before. However, the security protocol used is straightforward [31], [32]. The client sent the fingerprint and RFID directly to the verifier. This mechanism is vulnerable to sniffing attacks. Also, the previous research is not considered mutual authentication [31], [32]. The absence of mutual authentication is vulnerable to a impersonate attack.

Furthermore, this research proposed two-factor authentication using RFID and fingerprint to conduct secure mutual authentication—the RFID used as the first factor, and fingerprint as the second factor. The three stages that conduct mutual authentication are handshaking, proofing, and verifying. The handshaking stage is used to request the verifier to begin the authentication process. The proofing stage is used to evaluate that the client is legitimate. Furthermore, the verifying stage is used to validate that the client communicates with a legitimate verifier. Furthermore, the fingerprint is asses to find the threshold to verify the user. The brute force attack was conducted to evaluate the security by sending random properties. Next, the sniffing attack was conducted to evaluate the message sent through the network.

II. PREVIOUS WORK

Previous research has been conducting to provide security in IoT devices. Previous research has enabled two-factor authentication by using fingerprint and RFID as factors. Research conducted by Daulay et al. proposed a method to secure the door using fingerprint and RFID [33]. The proposed method was done using ATmega328P. The proposed method succeeded in authenticating either fingerprint and RFID. However, the research did not consider two-factor authentication.

The research was conducted in [34] trying to use SHA-3 to provide security in a microcontroller. The SHA-3 method for security using the Arduino Mega 2560 has resistance to key length extension and collision attacks compared to SHA-2. However, the SHA-3 method requires more computation time. SHA-256, a family of SHA-2, also requires low-cost arithmetic logic such as addition, shifts, rotation, and XOR. The research conducted by Wang [35] evaluated the algorithm between SHA-1, SHA-2, and SHA3.

There are other threats, such as collision attacks in RFID and fingerprint. The research conducted in [36] experimented with collision attacks on RFID using the Tav128 method. There were two collisions from 2^{32} inputs; collisions obtained from 2^{26} computations. This further means the Tav-128 hash function is not resistant to collision attacks.

The research was conducted in [37] discussing security on IoT devices connected to the internet. Node tampering attacks often occur on IoT devices that store data intact on EEPROM storage. Node tampering is a physical attack that could be successful in retrieving data. The attack aims to take over control of the device. Replay attacks often happen as an intrusion or unauthorized use, resulting in data - data can be stolen and misused or can change the information of the original sender. Furthermore, the IoT device must have

an identity from the sender or the receiver to prevent replay attacks.

III. METHOD

Two-factor authentication is one way to reduce the danger of password cracking [19]. Two-factor authentication uses a multi-layered identification process such as the use of something you have (RFID card) and something you are (fingerprint), both of which are used to identify a person in the first stage by tapping the RFID card and the second stage by tapping the fingerprint. If both stages are successful but identification fails, the system will reject it, but if both stages are successful and identification is successful, the system will accept it.

A. General Communication Model

The system consists of a client, an MQTT broker, and an authentication server. The client used ESP32. Then, the MQTT broker exchanges messages between the client and the authentication server. The MQTT broker used is publicly available. Then, the authentication server used is the Raspberry Pi. The Raspberry Pi has enough computing to run the proposed authentication protocol. The authentication client and server communicate with the MQTT broker simultaneously. The message topic for communicating with the authentication server has been defined. The "authentication" topic is used to carry out the authentication process, and the "registration" topic is used to carry out the registration process. Client and server authentication subscribe to both topics. The system built is shown in Figure 1.

The client is tasked with obtaining RFID and fingerprint data in the environment. The client has two roles: registering RFID and fingerprint data to the authentication server and carrying out the authentication process for RFID and fingerprint data to the authentication server. Then, the authentication server plays a role in executing, storing, and validating the RFID and fingerprint data. The authentication server accepts two types of requests, registration and authentication requests. The registration request is storing the RFID and fingerprint data to storage, and the authentication request is the process of validating the RFID and fingerprint data received from the client. MQTT broker acts as a place to exchange messages between the client and the authentication server. MQTT broker exchanges messages based on the topic of the messages it receives. Two processes are proposed to achieve the process defined above: the registration process and the authentication process. The registration process is registering RFID and fingerprints to the authentication server. Then, the authentication process is an RFID and fingerprint validation process sent by the client.

B. Registration process

The registration process is carried out once and is carried out before the authentication process occurs. The registration process is done in controlled environment. The registration process starts with the client sending a registration request

to the authentication server. Then, the authentication server send the public key $PubKy$ to the client. Next, the client is instructed to read the RFID data. The client computes the hash for the retrieved RFID data as $hrfid$ when the read is complete. Then, the client instructs to read the fingerprint. If the fingerprint reading is valid, the client calculates the fingerprint hash as hfp . Next, the RFID hash and the fingerprint hash are encoded using the authentication server's public key $PubKy$ that the client has received. Next, the client sends the encrypted hash $Reg1$ of the RFID and the fingerprint's hash $Reg2$ to the authentication server. The authentication server decrypts the message received from the client for registration with the private key $PrivKy$. Then, the authentication server stores the RFID hash and the fingerprint hash in its storage. Next, if the save process is successful, the authentication server sends an OK message to the client. The entire registration process protocol is shown in Figure 2.

C. Authentication process

The authentication process is carried out after the registration process has been completed. The authentication process begins with the client sending an authentication request message to the authentication server. The authentication server replies to the authentication request message by sending the authentication server's public key $PubKy$. Next, the client calculates a random number Rn . The random number element is essential in the key exchange and authentication processes. Then, the client encodes the random number Rn obtained with the public key $PubKy$ belonging to the authentication server. Next, the client is instructed to read the RFID data. The client calculates the hash of the RFID data $hrfid$ after the RFID data is read validly. Then, the client encodes the RFID data hash with a random number Rn symmetric key become $Auth1$. The client is instructed to read the fingerprint data. The client calculates the fingerprint's hash hfp after the fingerprint data is read validly with a predetermined threshold. Next, the client encodes the hash of the fingerprint data with a symmetric key random number Rn become $Auth2$.

The client sends the random number encoding $KeyX$, RFID data hash $hrfid$, and fingerprint data hash hfp to the au-

thentication server. The authentication server decrypts the random number Rn with its private key $PrivKy$. Then, the authentication server decrypts the RFID data hash $hrfid$ and the fingerprint data hash hfp using the previously obtained random number Rn . Then, the authentication server validates the hash of the RFID and fingerprint data with those stored. If the validation is declared granted, the authentication server calculates a random $sessionID$. Then, $sessionID$ is encoded with a symmetric key random number Rn . Then, the authentication server encodes the random number Rn with a symmetric key random number Rn -this is done to inform the client about mutual authentication. The authentication server sends the encoding $sessionID$ as $KeySv$ and random number encoding as $Auth3$ to the client. The client decrypts the random number Rn sent by the authentication server with its symmetric key random number Rn . If the random number sent by the authentication server is the same as the random number owned by the client, the client authenticates the message. Next, the client decrypts the $KeySv$ message with a random number Rn symmetric key. Furthermore, the client uses the $sessionID$ symmetric key to communicate with other clients on the

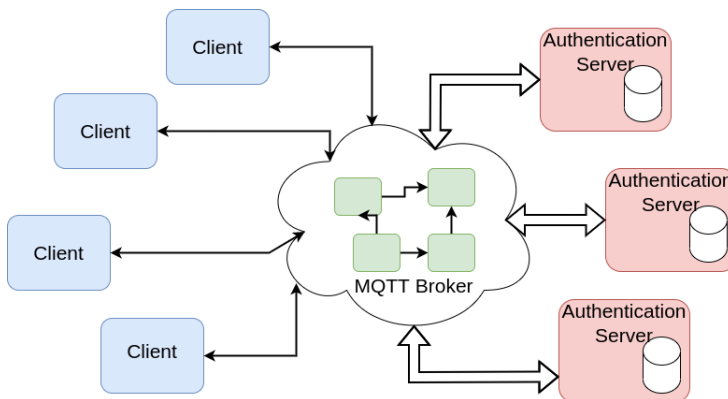


Fig. 1. General communication model

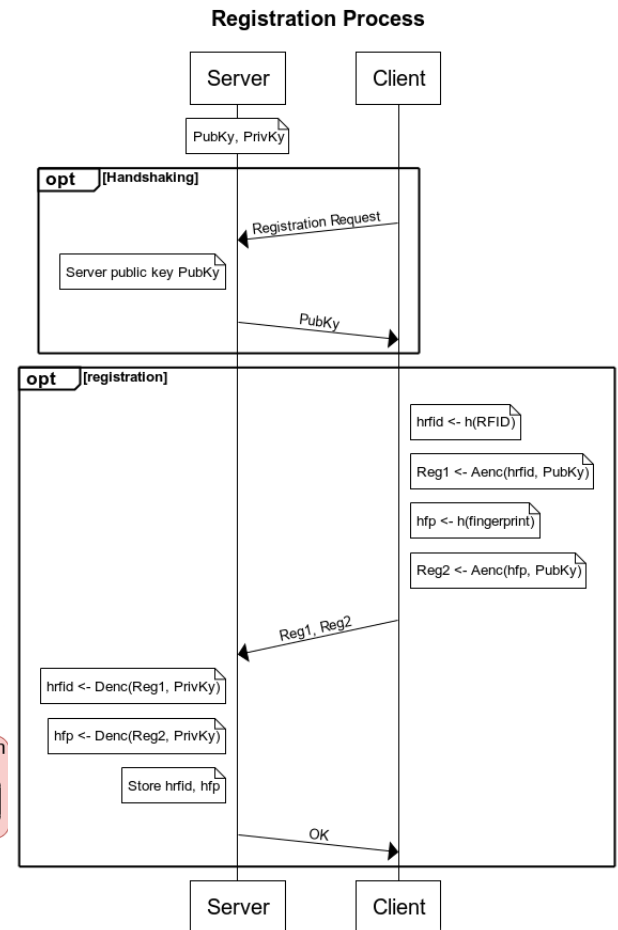


Fig. 2. Registration process

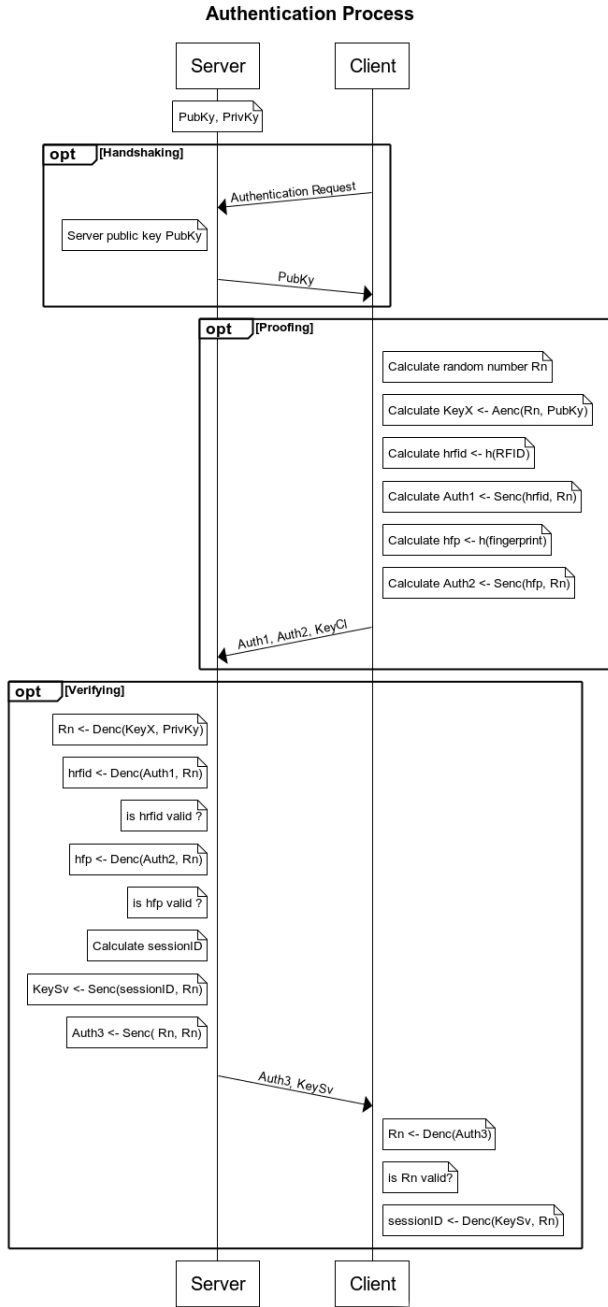


Fig. 3. Authentication process

network. The whole process of the authentication protocol is shown in Figure 3.

IV. EVALUATION AND DISCUSSION

A. Biometric Evaluation

The biometric evaluation test was conducted 60 times. The evaluation included 30 times testing the correct data to test the false acceptance rate (FAR) and 30 times testing on the wrong data to test the false rejection rate (FRR). Furthermore, the evaluation used fingerprint threshold of 80%, 70%, 60%, 50%,

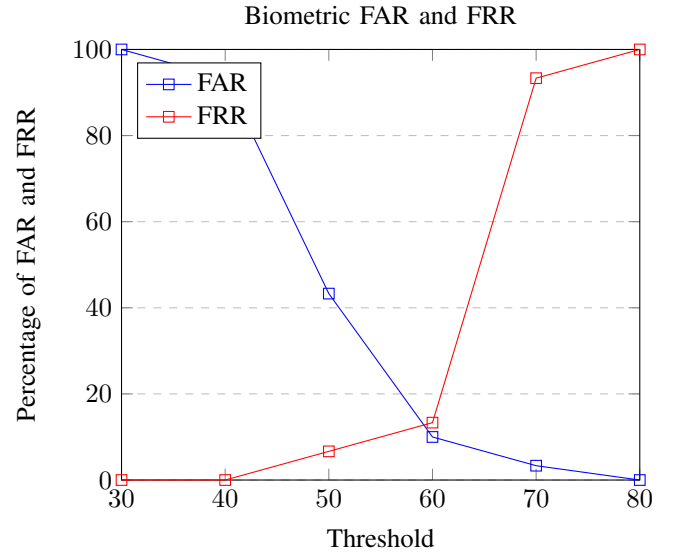


Fig. 4. Biometric FAR and FRR

40%, and 30%. The threshold used is the value of the number of similarities of each fingerprint data. The length of the data from the fingerprint is 1024 bits, so the test to be carried out is to calculate the value of False Acceptance Rate (FAR) and False Rejection Rate (FRR). So that the appropriate threshold value for use in the system used will show its accuracy. The FAR and FRR tests are shown in Figure 4.

The FAR and FRR tests are shown in Figure 4. Figure 4 shows that each threshold gives a different FAR and FRR. Of course, the desired goal is to minimize the FRR as much as possible and increase the FAR as high as possible. It can be seen in Figure 4 that the higher the threshold, the lower the FAR value, and the higher the FRR. The optimal value of FAR and FRR is at the threshold of 80%, where the FAR value is 0%, and FRR is 100%. Thus, at the 80% threshold, all users who do not have access rights are not granted access to the system, and all users who have access rights are granted access to the system. Furthermore, the equal error rate value obtained at the threshold is around 59.5%. In general, the lower ERR, the higher accuracy is. 59.5% state that the threshold should be set for the FAR higher than 59.5%.

B. Security Evaluation

1) *Brute Force Attack*: Brute force attack testing is intended to test the robustness of the authentication protocol built. Testing is done by randomly sending fingerprint and RFID data to the authentication server. This brute force aims to gain access to the authentication server. In the brute force test, the experiment was carried out 50 times. Each experiment was carried out as many as 65,536. Thus, a total of 3,276,800 tests were carried out. The brute force attack is shown in Figure 5. The experimental results found that the brute force attack did not succeed in gaining access to the authentication server. The failure was due to each authentication session always getting a

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ead18099c4dcd6ace8e7af21ee6f99500f290dc707ecb1fa429597cc6189da4ea5c147da7e1a82ee4a0fdd72f154ad5ebc
a501e758d05269f2788d43326671e76903719b82ba5192f4b872c7393da5c1793c3174854c7a2b555ac767d3f4c2fcb1ed
bf2f1b1f508ddfc3ad2ed7ec4533fdebbd5e9c97d7120e437d07d21fde0ecf220cae582e7d43467f10f408939c4a0a4
337d930eac0c5e4620a0c1d29a8e4f9f06b8f3f1e81b8cb0ab5f1e90c181b1ebac3105b8417b361901fcd7878ab472d40
31463855f81682d1d493d6946e3abb51abb70d51597d33b71701bc260389a000bf4f0fdd00034e910ec57855e94dc
304da3bade317634baf1469599e2ad59c7dc76bb1285cb2039c2d6f5e89bb50910c1d5da57e47be7dc2e7f7edbb27391
e42d7eb5152a62ef9d7fa21f55a62d202d6dc0265f1f9a9a43be6d1a0d5d9f02fcc4cabafbac1eeebd481c75df36c8aa
982c1edd04fe186d8ba8bb28635249b38efe1d75829b777592eded2c3a6f9bdf316323a839441998197a138fe832e637f7f9
a48d0bee79a6f74bf9a9b001264744
Hash RFID : 59f2527a993dab2c9977d7a15257e387443e5715d6358ad89eb7671cddb44
Gagal
Waktu : 29.616 detik.
Hash Fingerprint : ab313fe09b5b7860a2b72abebc752170990e115eb167aa7795e3ee234a80692106cebb83128dd4a
9055d8ce9af7c0e74318ba4380e225a809f87ce36e509a702d705eb9b73d6333d229770217f868cd0977f1c1353
7ee88ba76c55e5a702488ebf5518d3faef75cd415ee88dd2fcb5b3c872d2c590945ab9619fbc47cc3f8d6fb109c07a0
47392c06d188fc1ce558f9f52a3cd2a11cd4b3f970cf6c333d208a76c5dd29531fb1e92803134958205d5c35eb8ff616
bc360b13571311cc9137d01d56b06ed3f1f35712cc96172854523bf86ab1a44cbd4c4ed9113aba4329f9a8cf36d421b44
1fa22b074b711644e387c98be9b7f7d3b37477b0d4b48f90424abbfd910ce5108e4141fead67ea4c9462dae3246bdc4b01
04abb48b781dd4e3794151aa99499f7c77d23732edebd189550b12e591d9f7c7316c52ff188f3fcd6de9e9e06d0e71
00e0e0c95d31064f4d9595a88063fced5d2d4e9ba8468b95ed1e8bce1b04b673c8ce46f4c4c9a29576711a0e42f094
2881310e69326872c20b8ce8eb8bc1df6584bc3379a8a1a1f7b9164fb3baef351cdc0afdc3f393dd5f3f5b3311a841da68
44d6835586b5162b36b65a13ab1a95411e68b4c96aeb381add4c3e7e6a1fe06d5d02e5a23f572a2a1ed181cbde0f058b
0fe4a14290735879bdcab1fa82681
Hash RFID : 558d22882a78edf9b5ca38a25a4ff0cc44cc0cf178a22452f895dc763538
Gagal

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Fig. 5. Brute force attack

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Topic: authSes_id
Message: 72

0000 4f 50 4f 56 2b 31 47 64 6a 6d 30 0e 00 0a 61 75 OPOV+1gd jmm0...au
00d0 74 68 53 65 73 5f 69 64 37 32 thSes_id 72

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Fig. 6. Sniffing attack on sessionID

different random number. The different random number causes the data sent in each session to change constantly.

2) *Sniffing Attack*: Sniffing attacks are intended to test messages sent by clients over the MQTT network. This attack extracts messages sent by intercepting messages between the client and the authentication server. The sniffing scenario is done by capturing packets using Wireshark for sessionID, RFID, and fingerprint data taken from communication between the client and server using the MQTT protocol. First, the sniffing attack aims to obtain the sessionID sent by the client to the authentication server. sessionID is essential because it is a symmetric key. Sniffing results on sessionID are shown in Figure 6. The client sent sessionID is 18. The sessionID is then encoded with RSA to become 72. It can be seen that the value sent by the client has been encrypted using the RSA algorithm. Second, the sniffing attack aims to obtain RFID data sent by the client to the authentication server. RFID data is crucial because it is one of the two factors in the authentication process. The results of sniffing on RFID data are shown in Figure 7. RFID ciphertext data sent by the client has been encrypted with a symmetric key from the sessionID value. Third, the sniffing attack is intended to obtain the client's fingerprint data sent to the authentication server. Fingerprint data is one of two authentication factors. The sniffing results from fingerprint data are shown in Figure 8. The ciphertext fingerprint data sent by the client has been encrypted with a symmetric key from the sessionID value.

V. CONCLUSION

The development of IoT has penetrated various sectors. The development of IoT devices continues to increase and is predicted to reach 75 billion by 2025. However, the development of IoT devices is not followed by security developments. IoT devices can become a new gateway for cyber attacks. One of the attacks on IoT devices is a brute force attack and sniffing. There needs a way to ward off cyber attacks. Authentication mechanisms can be used to ward off attacks.

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Topic: authRFid
Message: rvZmo41/h99n0cvz+LTAq2ho1d0P+Zmq/A1BHRGyC13mUuY4D1bZ1E+U

0000 94 e9 79 fe c0 cd 24 6f 28 a0 de 04 08 00 45 00 y...$o (...E
0010 00 b5 00 09 00 ff 06 85 2a c0 a8 5a 3d c0 a8 Z...:..*..Z=
0020 5a 81 d9 f8 07 5b 00 00 19 d8 21 05 76 13 50 18 Z...[...v P
0030 16 a5 a2 8a 00 00 30 8a 01 00 08 61 75 74 68 52 E...0...authR
0040 66 69 64 72 76 5a 6d 6f 34 6c 2f 68 39 39 6e 44 fidrvZmo 41/h99n0
0050 38 7f 07 80 2b 38 4c 54 41 71 32 68 6f 31 64 30 6f 05a+LTA q2ho1d0e
0060 50 2b 5a 6d 51 2f 41 6c 42 48 4e 52 67 79 43 69 +Zmq/A1 BHRGyC1
0070 33 6d 55 75 59 34 44 69 62 49 5a 6c 45 2b 55 6b 3mUuY4D1 bZ1E+Uk
0080 32 5a 4b 47 6f 35 52 4e 6e 51 78 63 35 6e 30 4e 2ZKGo5RN nQxc5n0N
0090 34 46 65 65 62 78 49 77 50 6c 68 41 57 45 73 58 4FeebIw P1jAwEsX
00a0 73 76 76 5a 69 56 42 37 69 78 6c 6a 61 79 57 69 sNvZ1VB7 1x1jAwW1
00b0 65 5a 43 5a 52 34 37 36 77 61 78 6d 66 48 59 6a eZCZr476 waxmFHP
00c0 37 35 58 75X

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Fig. 7. Sniffing attack on RFID

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MQ Telemetry Transport Protocol: Publish Message
Header Flags: 0x30, Message Type: Publish Message, QoS Level: At most once delivery (Fire and Forget)
Msg Len: 146
Topic Length: 16
Topic: authFingerprint
Message: 81c1Mag12xth0K10f1e49Y0Ly57BHL0Ch0xt0tXdrtsA1K5V6/qhZr1f2Gw6/H6I/mYEN3E1b5TPWcdtGofg8y/gb0zSPORC1

MQ Telemetry Transport Protocol: Publish Message
Header Flags: 0x30, Message Type: Publish Message, QoS Level: At most once delivery (Fire and Forget)
Msg Len: 146
Topic Length: 16
Topic: authFingerprint2
Message: 81c1Mag12xth0K10f1e49Y0Ly57BHL0Ch0xt0tXdrtsA1K5V6/qhZr1f2Gw6/H6I/mYEN3E1b5TPWcdtGofg8y/gb0zSPORC1

MQ Telemetry Transport Protocol: Publish Message
Header Flags: 0x30, Message Type: Publish Message, QoS Level: At most once delivery (Fire and Forget)
Msg Len: 146
Topic Length: 16
Topic: authFingerprint
Message: 81c1Mag12xth0K10f1e49Y0Ly57BHL0Ch0xt0tXdrtsA1K5V6/qhZr1f2Gw6/H6I/mYEN3E1b5TPWcdtGofg8y/gb0zSPORC1

0000 68 46 69 6e 67 65 72 70 72 69 6e 74 31 c1 47 66 nFingerprint0
0010 69 68 36 30 49 75 31 51 52 30 4b 55 51 52 4a 39 nFingerprint1
0020 42 44 38 40 48 42 64 32 0f 79 47 53 49 40 40 6f nFingerprint2
0030 78 4c 49 72 50 51 43 67 63 6e 39 4a 6c 47 33 29 nFingerprint3
0040 65 76 71 42 58 50 50 69 77 44 65 60 40 6f 6f nFingerprint4
0050 57 44 37 31 50 54 7a 4b 47 75 46 39 44 50 56 69 nFingerprint5
0060 4c 47 6e 60 76 63 6c 4a 71 6c 68 49 60 6f 6f nFingerprint6
0070 47 59 35 46 52 41 36 46 59 18 90 90 91 nFingerprint7
0080 00 10 61 75 74 68 68 69 6e 67 65 72 70 72 69 6e nFingerprint8
0090 74 32 38 69 63 40 61 6f 47 63 6c 39 32 60 74 6b nFingerprint9
0100 73 64 4b 33 4c 4f 46 62 69 65 34 39 45 79 49 6f nFingerprint10
0110 4c 73 59 35 37 42 48 4c 4f 43 68 39 74 74 6f 6f nFingerprint11
0120 74 58 44 72 74 73 4b 41 31 56 53 47 27 63 51 68 nFingerprint12
0130 5a 72 49 66 5a 47 71 77 30 2f 48 36 49 2f 60 79 nFingerprint13

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Fig. 8. Sniffing attack on fingerprint

A secure authentication mechanism can guarantee secure communications and data. However, the implementation of authentication mechanisms on IoT devices is challenging. IoT devices are dominated by constraint devices that have limited computing. The use of conventional authentication mechanisms becomes unsuitable for use. Two-factor authentication using RFID and fingerprint can be a solution in providing an authentication mechanism on IoT constraint devices. Previous studies have proposed a two-factor authentication mechanism using RFID and fingerprint. However, previous research did not pay attention to security as a whole. Previous research did not pay attention to the security of data transmission on the network. In addition, previous studies did not provide a mutual authentication mechanism. This study proposes a secure mutual authentication protocol using two-factor RFID and fingerprinting using the MQTT protocol. Two processes support the authentication process: the registration process and authentication. The registration process aims to securely register RFID and fingerprint data from the client to the authentication server. Then, the authentication process aims to validate the RFID and fingerprint data from the client securely.

The proposed protocol is tested based on biometric security by measuring the false acceptance rate (FAR) and false rejection rate (FRR) on the fingerprint, brute force attacks, and sniffing attacks. The test results obtained the most optimal FAR and FRR at the 80% threshold. Then the equal error rate (ERR) on FAR and FRR is around 59.5%. Then, brute force attack testing found that the proposed protocol was resistant to these attacks. Brute-force attacks can be prevented by using a dynamic random number symmetric key. Then, testing sniffing attacks found that the proposed protocol is resistant to these attacks. Sniffing attacks can be anticipated because the sent RFID and fingerprint data are already encrypted.

Physical attacks are the most common attacks aimed at IoT devices. Testing will be carried out against physical attacks such as cloning in future research. Furthermore, methods will be investigated to obtain a secure and proper random number, such as using a physical unclonable function (PUF). Random sources are essential in providing a secure authentication mechanism.

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